

B.Tech III Year I Semester (R20) Regular Examinations January 2023

MACHINE LEARNING

(Common to AI&DS, CSE(AI&ML) & CSE(AI))

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

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| (a) What the importance of Inductive Bias. | 2M |
| (b) Differentiate training Data and testing Data. | 2M |
| (c) What is the purpose of pruning? | 2M |
| (d) How to avoid overfitting of data? | 2M |
| (e) Define Vapnik-Chervonenkis Dimension. | 2M |
| (f) What is PAC learning? | 2M |
| (g) What is meant by gradient descent? | 2M |
| (h) What is the purpose of multi-layer networks? | 2M |
| (i) State the advantages of instance-based learning. | 2M |
| (j) What is meant by logistic regression? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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| 2 (a) Explain different perspectives and issues in machine learning. | 5M |
| (b) What are the concepts of learning as search? | 5M |

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| 3 Explain Candidate-elimination learning algorithm using version spaces. | 10M |
| 4 Explain about decision tree-based learning. How it is represented? Describe some problems for which decision tree-based learning is appropriate. | 10M |

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| 5 Describe the following:
(i) Entropy. (ii) Information Gain. (iii) Gain Ratio. | 10M |
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| 6 Discuss about first order rule learning in detail. | 10M |
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| 7 (a) Explain about Inverting resolution. | 5M |
| (b) Describe models of learnability. | 5M |

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| 8 Explain in brief about Back propagation algorithm. | 10M |
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| 9 Explain Quadratic programming solution for finding maximum margin separators. | 10M |
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| 10 Differentiate generative and discriminative training in detail. | 10M |
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OR

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| 11 What are Bayesian belief nets where are they used in machine learning? | 10M |
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